

REKURZIJE: Dodatne vaje

1. $a_n - 5a_{n-1} + 6a_{n-2} = 0, a_0 = 2, a_1 = 4$

2. $a_n - 10a_{n-1} + 25a_{n-2} = 0, a_0 = 3, a_1 = 17$

3. $a_n - 2a_{n-1} + 2a_{n-2} = 0, a_0 = 1, a_1 = 3$

4. $a_{n+2} - 2a_{n+1} + a_n = 3n + 1$

5. $a_n - 3a_{n-1} - 10a_{n-2} = 7 \cdot 5^n$

Rešitve:

1. $a_n = A \cdot 3^n + B \cdot 2^n$

2. $a_n = (3 + \frac{2}{5}n) \cdot 5^n$

3. $a_n = (\sqrt{2})^n (\cos \frac{\pi n}{4} + 2 \sin \frac{\pi n}{4})$

4. $a_n = A + Bn - n^2 + \frac{1}{2}n^3$

5. $a_n = 6 \cdot (-2)^n - 2 \cdot 5^n + 5n \cdot 5^n$